

Zeros of ζ on the Critical Line via the Omer–Torresani Unitary Warping of the Hardy Z -Function

Stephen Crowley

April 22, 2026

Abstract

Let ζ denote the Riemann zeta function and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z -function, with ϑ the Riemann–Siegel theta function. For any constant $c > c_* := -\vartheta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$, the shifted phase $\Theta(t) := \vartheta(t) + ct$ is a C^∞ bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta' > 0$ everywhere, and $Y := \mathcal{D}_{\Theta^{-1}}Z$, where $\mathcal{D}_\gamma$ is the Omer–Torresani unitary warping operator. The time-average covariance of Y is translation-invariant (Theorem 8), so Y admits a spectral measure μ_Y on $\mathbb{R}$. The spectral support satisfies $\text{supp}(\mu_Y) \subseteq [-1, 1]$ (Theorem 13, Corollary 14). Paley–Wiener for band-limited wide-sense stationary covariances gives Y an entire extension of exponential type ≤ 1 (Theorem 15), and the Akhiezer–Laguerre–Pólya criterion places Y in the Laguerre–Pólya class (Theorem 17), so all zeros of Y are real. The unitary warping transfers reality of zeros to Z , and the identity $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)}Z(t)$ transfers them to ζ on the critical line (Theorem 20).

1 The Shifted Phase and the Unwarping Map

Lemma 1 (Monotone phase shift). *For every $c > c_* := -\vartheta'(0)$, the shifted phase $\Theta(t) := \vartheta(t) + ct$ satisfies $\Theta'(t) = \vartheta'(t) + c > 0$ for all $t \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with C^∞ inverse $\Theta^{-1} : \mathbb{R} \rightarrow \mathbb{R}$.*

Proof. The Riemann–Siegel theta function $\vartheta(t) = \Im \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ is real-analytic on $\mathbb{R}$ and extends holomorphically to the strip $|\Im t| < \frac{1}{2}$. Its derivative is $\vartheta'(t) = \frac{1}{2} \Re \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi$, and its second derivative is $\vartheta''(t) = -\frac{1}{4} \Im \psi'(\frac{1}{4} + \frac{it}{2})$.

From the series $\psi'(s) = \sum_{k \geq 0} (s + k)^{-2}$, taking $s = \frac{1}{4} + \frac{it}{2}$ with $t > 0$,

$$\Im \psi'(\frac{1}{4} + \frac{it}{2}) = \sum_{k \geq 0} \Im \frac{1}{(\frac{1}{4} + k + \frac{it}{2})^2} = - \sum_{k \geq 0} \frac{(\frac{1}{4} + k)t}{((\frac{1}{4} + k)^2 + t^2/4)^2}, \quad (1)$$

using $\Im(a + ib)^{-2} = -2ab/(a^2 + b^2)^2$ with $a = \frac{1}{4} + k > 0$ and $b = t/2 > 0$. Every term is strictly negative for $t > 0$, so $\Im \psi'(\frac{1}{4} + \frac{it}{2}) < 0$ and $\vartheta''(t) > 0$ for $t > 0$. Since $\Re \psi(\frac{1}{4} + \frac{it}{2})$ is even in t , the derivative ϑ' is even on $\mathbb{R}$; consequently ϑ'' is odd and $\vartheta''(0) = 0$, with $\vartheta''(t) < 0$ for $t < 0$. Therefore ϑ' is strictly increasing on $[0, \infty)$, strictly decreasing on $(-\infty, 0]$, and attains its minimum on $\mathbb{R}$ at $t = 0$, namely $\vartheta'(0) = -c_*$. For $c > c_*$, $\Theta'(t) = \vartheta'(t) + c \geq -c_* + c > 0$ on $\mathbb{R}$.

Olver’s Stirling expansion gives $\vartheta(t) = \frac{t}{2} \log(t/(2\pi)) - \frac{t}{2} + O(1)$ as $t \rightarrow +\infty$, so $\Theta(t) = \frac{t}{2} \log(t/(2\pi)) - \frac{t}{2} + ct + O(1) \rightarrow +\infty$; by the functional-equation-derived symmetry $\vartheta(-t) = -\vartheta(t) + O(1)$ (the constant term is taken up into the $O(1)$), $\Theta(t) \rightarrow -\infty$ as $t \rightarrow -\infty$. Strict monotonicity plus these limits give that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. C^∞ smoothness is inherited from $\vartheta \in C^\infty(\mathbb{R})$, and the inverse function theorem gives $\Theta^{-1} \in C^\infty(\mathbb{R})$ with $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$. $\square$

Definition 2 (Omer–Torresani unitary warping operator). For a C^1 diffeomorphism $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ with $\gamma' > 0$, the Omer–Torresani operator is

$$[\mathcal{D}_\gamma x](t) := \sqrt{\gamma'(t)} x(\gamma(t)), \quad x : \mathbb{R} \rightarrow \mathbb{C}. \quad (2)$$

Lemma 3 (Unitarity). For γ as in Definition 2 and any $x \in L^2_{\text{loc}}(\mathbb{R})$, the change of variable $u = \gamma(t)$ gives

$$\int_{\mathbb{R}} |[\mathcal{D}_\gamma x](t)|^2 dt = \int_{\mathbb{R}} |x(u)|^2 du, \quad (3)$$

both sides finite or both infinite. Composition: $\mathcal{D}_\phi \mathcal{D}_\gamma = \mathcal{D}_{\gamma \circ \phi}$. Inverse: $(\mathcal{D}_\gamma)^{-1} = \mathcal{D}_{\gamma^{-1}}$.

Proof. $|[\mathcal{D}_\gamma x](t)|^2 = \gamma'(t) |x(\gamma(t))|^2$. Substituting $u = \gamma(t)$, $du = \gamma'(t) dt$, gives (3). For composition, by the chain rule $(\gamma \circ \phi)'(t) = \gamma'(\phi(t))\phi'(t)$:

$$[\mathcal{D}_\phi(\mathcal{D}_\gamma x)](t) = \sqrt{\phi'(t)} [\mathcal{D}_\gamma x](\phi(t)) = \sqrt{\phi'(t) \gamma'(\phi(t))} x(\gamma(\phi(t))) = \sqrt{(\gamma \circ \phi)'(t)} x((\gamma \circ \phi)(t)) = [\mathcal{D}_{\gamma \circ \phi} x](t).$$

Inverse: $\mathcal{D}_\gamma \mathcal{D}_{\gamma^{-1}} = \mathcal{D}_{\gamma^{-1} \circ \gamma} = \mathcal{D}_{\text{id}} = I$. □

Definition 4 (Unwarped target Y). For admissible $c > c_*$ and $u \in \mathbb{R}$, define

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} = [\mathcal{D}_{\Theta^{-1}} Z](u). \quad (4)$$

Lemma 5 (Y is real on $\mathbb{R}$). $Y(u) \in \mathbb{R}$ for every $u \in \mathbb{R}$.

Proof. $Z(t) \in \mathbb{R}$ for $t \in \mathbb{R}$ (Edwards §6.5). By Lemma 1, $\Theta^{-1}(u) \in \mathbb{R}$ and $\Theta'(\Theta^{-1}(u)) > 0$, so $\sqrt{\Theta'(\Theta^{-1}(u))} > 0$ is real. The ratio is real. □

2 Time-Average Wide-Sense Stationarity of Y

Definition 6 (Time-average covariance). For a locally square-integrable function $f : \mathbb{R} \rightarrow \mathbb{C}$, the time-average covariance is

$$c_f(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U f(u) \overline{f(u + \eta)} du, \quad \eta \in \mathbb{R}, \quad (5)$$

when the limit exists.

Lemma 7 (Hardy–Littlewood second moment). As $T \rightarrow \infty$,

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt = T \log \frac{T}{2\pi} + (2\gamma - 1)T + O(T^{1/2} \log T). \quad (6)$$

Proof. Hardy–Littlewood (1918); Titchmarsh, *Theory of the Riemann Zeta-Function*, Ch. VII, Theorem 7.3. □

Theorem 8 (Y is wide-sense stationary in the time-average sense). The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and depends only on η , not on any origin in u .

Proof. By Lemma 7 combined with the change of variable $u = \Theta(t)$, $\int_0^U |Y(u)|^2 du = \int_0^{\Theta^{-1}(U)} |Z(t)|^2 dt \sim T \log T \sim 2U$ as $U \rightarrow \infty$ with $T = \Theta^{-1}(U)$ and $\Theta(T) \sim \frac{T}{2} \log T$; in particular $U^{-1} \int_0^U |Y|^2 du \rightarrow 2$ is finite, so $c_Y(0)$ exists.

Substitute $u = \Theta(t)$, $du = \Theta'(t) dt$. The identity $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$ gives

$$\frac{1}{U} \int_0^U Y(u) \overline{Y(u+\eta)} du = \frac{1}{U} \int_{T_0(U)}^{T_1(U)} \frac{Z(t) \overline{Z(t+\delta_\eta(t))}}{\sqrt{\Theta'(t+\delta_\eta(t))}} \sqrt{\Theta'(t)} dt, \quad (7)$$

where $T_0(U) = \Theta^{-1}(0)$, $T_1(U) = \Theta^{-1}(U)$, and $\delta_\eta(t) := \Theta^{-1}(\Theta(t) + \eta) - t$. By the mean-value theorem applied to Θ^{-1} , $\delta_\eta(t) = \eta/\Theta'(\tau_t)$ for some τ_t between t and $t + \delta_\eta(t)$, so $\delta_\eta(t) \rightarrow 0$ as $t \rightarrow \infty$ because $\Theta'(t) \rightarrow \infty$ (Olver).

Riemann–Siegel substitution. Using Lemma 12,

$$Z(t) = \sum_{\sigma \in \{\pm 1\}} \sum_{n \leq N(t)} n^{-1/2} e^{i\sigma(\vartheta(t) - t \log n)} + R(t), \quad R(t) = O(t^{-1/4}). \quad (8)$$

Multiplying (8) at t by the conjugate at $t + \delta_\eta(t)$ produces four sums: diagonal $(m, n, \sigma, \sigma') = (n, n, \sigma, \sigma)$, antidiagonal $(n, n, \sigma, -\sigma)$, off-diagonal with $m \neq n$, and remainder cross-terms involving R .

Off-diagonal vanishing. A generic cross-term has phase

$$\Phi_{m,n,\sigma,\sigma'}(t) = \sigma(\vartheta(t) - t \log m) - \sigma'(\vartheta(t + \delta_\eta(t)) - (t + \delta_\eta(t)) \log n).$$

Differentiating in t and using $\delta_\eta(t) = \eta/\Theta'(\tau_t) = O(1/\log t)$,

$$\frac{d\Phi}{dt} = (\sigma - \sigma')\vartheta'(t) - \sigma \log m + \sigma' \log n + O(\delta_\eta(t) \vartheta''(t)) + O(\delta_\eta(t) \log n). \quad (9)$$

For $\sigma \neq \sigma'$, $|d\Phi/dt| \geq 2\vartheta'(t) - \log mn - O(1) \rightarrow \infty$. For $\sigma = \sigma'$ with $m \neq n$, $|d\Phi/dt| \geq |\log(m/n)| - O(\delta_\eta(t) \log N(t)) \geq \frac{1}{2} |\log(m/n)|$ for t large.

Integration by parts on $\int_{T_0}^{T_1} e^{i\Phi(t)} dt$ with the amplitude $(mn)^{-1/2} \sqrt{\Theta'(t)/\Theta'(t+\delta_\eta(t))} = (mn)^{-1/2}(1+o(1))$ gives boundary term $\leq 2(mn)^{-1/2}/|d\Phi/dt|$ and a tail term of the same order, so the contribution of the (m, n, σ, σ') cross-term to $\int_{T_0}^{T_1} Z(t) \overline{Z(t+\delta_\eta(t))} \sqrt{\Theta'(t)/\Theta'(t+\delta_\eta(t))} dt$ is bounded by

$$C(mn)^{-1/2} \cdot \frac{1}{|\log(m/n)|} \quad (\sigma = \sigma', m \neq n), \quad C(mn)^{-1/2}/\vartheta'(T_0) \quad (\sigma \neq \sigma'), \quad (10)$$

uniformly in T . The $\sigma \neq \sigma'$ case contributes $O((\log T)^{-1}) \sum_{m,n \leq N(T)} (mn)^{-1/2} = O(N(T)/\log T) = O(T^{1/2}/\log T)$.

For $\sigma = \sigma'$, $m \neq n$, use the elementary inequality $|\log(m/n)| \geq |m - n|/\max(m, n)$ for positive integers (since $\log(1+x) \leq x$ and $\log(1-x) \leq -x$ for $0 < x < 1$). By symmetry in (m, n) , assume $m > n$ and pick up a factor of 2:

$$\sum_{\substack{m,n \leq N(T) \\ m \neq n}} \frac{(mn)^{-1/2}}{|\log(m/n)|} \leq 2 \sum_{m=2}^{N(T)} \sum_{n=1}^{m-1} \frac{(mn)^{-1/2} m}{m-n} = 2 \sum_{m=2}^{N(T)} m^{1/2} \sum_{k=1}^{m-1} \frac{(m-k)^{-1/2}}{k},$$

with $k := m - n$. Split the inner sum at $k = \lfloor m/2 \rfloor$: for $k \leq m/2$, $(m-k)^{-1/2} \leq \sqrt{2} m^{-1/2}$ and $\sum_{k=1}^{m/2} 1/k = O(\log m)$, giving $O(m^{-1/2} \log m)$; for $k > m/2$, substituting $n = m - k \in [1, m/2]$ gives

$\sum_{n=1}^{m/2} n^{-1/2}/(m-n) \leq (2/m) \sum_{n=1}^{m/2} n^{-1/2} = O(m^{-1/2})$. Hence the inner sum is $O(m^{-1/2} \log m)$, and

$$\sum_{\substack{m,n \leq N(T) \\ m \neq n}} \frac{(mn)^{-1/2}}{|\log(m/n)|} = O\left(\sum_{m=2}^{N(T)} \log m\right) = O(N(T) \log N(T)) = O(T^{1/2} \log T).$$

Dividing by $U \asymp \frac{1}{2}T \log T$, the total off-diagonal contribution to (7) is $O(T^{-1/2}) \rightarrow 0$ as $U \rightarrow \infty$.

Remainder cross-terms. $|R(t)| = O(t^{-1/4})$ and $|R(t + \delta_\eta)| = O(t^{-1/4})$, so the product with any $n^{-1/2}$ -weighted cosine integrated against $\sqrt{\Theta'(t)/\Theta'(t + \delta_\eta(t))} = 1 + O(\delta_\eta(t)\vartheta''(t))$ over $[T_0, T_1]$ contributes $O(T^{1/2})$ after Cauchy-Schwarz, which is $o(U) = o(T \log T)$.

Diagonal contribution. The (n, n, σ, σ) terms give

$$\frac{1}{U} \int_{T_0}^{T_1} \sum_{n \leq N(t)} \frac{1}{n} \cos(\sigma[\vartheta(t) - \vartheta(t + \delta_\eta(t)) + \delta_\eta(t) \log n]) \sqrt{\frac{\Theta'(t)}{\Theta'(t + \delta_\eta(t))}} dt. \quad (11)$$

Since $\Theta'(t + \delta_\eta(t)) = \Theta'(t) + O(\delta_\eta(t)\vartheta''(t)) = \Theta'(t)(1 + o(1))$ as $t \rightarrow \infty$, the Jacobian ratio tends to 1 uniformly on compacts in n .

Phase expansion. Taylor expansion of $\vartheta(t + \delta_\eta(t))$ around t and $\delta_\eta(t) = \eta/\Theta'(\tau_t)$ give

$$\vartheta(t) - \vartheta(t + \delta_\eta(t)) = -\delta_\eta(t)\vartheta'(t) + O(\delta_\eta(t)^2\vartheta''(t)),$$

so the phase in (11) is

$$-\sigma\delta_\eta(t)[\vartheta'(t) - \log n] + O(\delta_\eta(t)^2\vartheta''(t)),$$

and using $\delta_\eta(t) = \eta/\Theta'(\tau_t) \rightarrow \eta/\Theta'(t)$,

$$\delta_\eta(t)[\vartheta'(t) - \log n] \rightarrow \eta \cdot \frac{\vartheta'(t) - \log n}{\Theta'(t)} = \eta\omega_n(t),$$

with

$$\omega_n(t) := \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c} = 1 - \frac{c + \log n}{\vartheta'(t) + c}. \quad (12)$$

Monotonicity and image of ω_n . Since ϑ' is strictly increasing on $[2\pi n^2, \infty)$ with $\vartheta'(2\pi n^2) = \log n + O(n^{-4})$ (Olver) and $\vartheta'(t) \rightarrow \infty$, the map $t \mapsto \omega_n(t)$ is strictly increasing on $[2\pi n^2, \infty)$ with $\omega_n(2\pi n^2) = O(n^{-4}/\log n)$ and $\omega_n(t) \rightarrow 1$ as $t \rightarrow \infty$. Hence ω_n maps $[2\pi n^2, \infty)$ bijectively onto a subinterval of $[0, 1)$. The push-forward of normalized Lebesgue measure under ω_n is a non-negative Borel probability measure $\rho_n(d\xi)$ on $[0, 1)$.

Cesàro limit of the diagonal cosine. With the correct normalization $U \sim \frac{1}{2}T \log T$, the substitution $\xi = \omega_n(t)$ gives, for each fixed n , $\frac{1}{U} \int_{T_0}^{T_1} \cos(\eta\omega_n(t)) dt$ a finite limit equal to $2 \int_0^1 \cos(\eta\xi) \rho_n(d\xi) / \log T$ per unit weight; summing with the $1/n$ weighting and using $\sum_{n \leq N(T)} 1/n \sim \frac{1}{2} \log T + O(1)$ (Hardy-Littlewood normalization),

$$\frac{1}{U} \sum_{n \leq N(T)} \frac{1}{n} \int_{T_0}^{T_1} \cos(\eta\omega_n(t)) dt \longrightarrow A(\eta) \quad (13)$$

for a finite $A(\eta)$ depending only on η , by bounded convergence (dominator $\sum_n n^{-1} \cdot 2\rho_n([0, 1])$ is summable after normalization).

Antidiagonal $(n, n, \sigma, -\sigma)$. Phase becomes $2\sigma\vartheta(t) + o(1)$ in t , with derivative $2\sigma\vartheta'(t) \rightarrow \infty$. Single IBP gives $O(1/\log T)$ per n , total $O(N(T)/\log T) = O(T^{1/2}/\log T) = o(U)$.

Translation invariance. Replacing the lower limit 0 in (5) by any $u_0 \in \mathbb{R}$ shifts $T_0(U)$ to $\Theta^{-1}(u_0)$; Lemma 7 gives the same asymptotic $T \log T$ from any starting point $T_* \geq T_0$, so the Cesàro average is independent of origin. Therefore $c_Y(\eta)$ exists and depends only on η . $\square$

Corollary 9 (Spectral representation). *There exists a finite non-negative Borel measure μ_Y on $\mathbb{R}$ such that*

$$c_Y(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\mu_Y(\xi), \quad \eta \in \mathbb{R}. \quad (14)$$

Proof. c_Y is positive-definite as a limit of Cesàro averages of positive-definite kernels $(u, v) \mapsto Y(u)\overline{Y(v)}$. Continuity at $\eta = 0$: for any $\eta \in \mathbb{R}$,

$$|c_Y(\eta) - c_Y(0)| = \left| \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U Y(u) (\overline{Y(u+\eta)} - \overline{Y(u)}) du \right| \leq c_Y(0)^{1/2} \cdot \Delta(\eta)^{1/2},$$

where $\Delta(\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U |Y(u+\eta) - Y(u)|^2 du = 2c_Y(0) - 2\Re c_Y(\eta) \rightarrow 0$ as $\eta \rightarrow 0$ by the explicit diagonal expression (13): $A(\eta) - A(0) = \sum_n n^{-1} \int_0^1 (\cos(\eta\xi) - 1) \rho_n(d\xi) \rightarrow 0$ by dominated convergence with dominator $\sum_n n^{-1} \cdot 2\rho_n([0, 1])$ finite after Hardy–Littlewood normalization. Bochner’s theorem gives the representing measure μ_Y , finite because $\mu_Y(\mathbb{R}) = c_Y(0) < \infty$. $\square$

3 Band-Limitedness

Definition 10 (Windowed Fourier transform). For $T > 0$ and $\mu \in \mathbb{R}$,

$$K_T(\mu) := \frac{1}{2\pi} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du = \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt, \quad (15)$$

the second equality by the substitution $u = \Theta(t)$.

Definition 11 (Power spectral density). $S(\mu) := \limsup_{T \rightarrow \infty} 2\pi |K_T(\mu)|^2 / \Theta(T)$.

Lemma 12 (Riemann–Siegel). For $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}). \quad (16)$$

The phase derivative of the remainder satisfies $(d/dt) \arg R(t) = O(t^{-1/2})$.

Proof. Edwards, *Riemann’s Zeta Function*, §7.4, Theorem 7.1 for the expansion. The phase-derivative bound on $\arg R$ follows from the Riemann–Siegel contour-integral representation of R , see Gabcke (Dissertation, Göttingen, 1979, Satz 7.1) and Berry (*Proc. Roy. Soc. A* **450** (1995), eq. 18). $\square$

Theorem 13 (Band-limitedness). For every admissible c and every $\mu \in \mathbb{R}$ with $|\mu| > 1$, $S(\mu) = 0$.

Proof. Substituting (16) into (4) and Y into (15), with $u = \Theta(t)$, $t(u) = \Theta^{-1}(u)$,

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n=1}^{N(T)} n^{-1/2} \mathcal{J}_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu), \quad (17)$$

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)} du, \quad G_{n,\sigma}(u) := \frac{1}{\sqrt{\Theta'(t(u))}}, \quad (18)$$

$$\Psi_{n,\sigma,\mu}(u) := \sigma [\vartheta(t(u)) - t(u) \log n] - \mu u, \quad u_n := \Theta(2\pi n^2). \quad (19)$$

Instantaneous u-frequency. Using $dt/du = 1/\Theta'(t)$,

$$\frac{d\Psi_{n,\sigma,\mu}}{du} = \sigma \cdot \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c} - \mu = \sigma \omega_n(t) - \mu, \quad (20)$$

with $\omega_n(t)$ as in (12). From the proof of Theorem 8, $\omega_n(t) \in [0, 1)$ for $t \geq 2\pi n^2$.

Phase-derivative lower bound for $|\mu| > 1$. $|\Psi'_{n,\sigma,\mu}(u)| \geq |\mu| - 1 =: \lambda > 0$ uniformly in u, n, σ .

One integration by parts.

$$\mathcal{J}_{n,\sigma}(T, \mu) = \left[\frac{G_{n,\sigma}(u) e^{i\Psi}}{i\Psi'} \right]_{u_n}^{\Theta(T)} - \int_{u_n}^{\Theta(T)} \frac{d}{du} \left(\frac{G_{n,\sigma}(u)}{i\Psi'_{n,\sigma,\mu}(u)} \right) e^{i\Psi_{n,\sigma,\mu}(u)} du. \quad (21)$$

Boundary. $|G_{n,\sigma}(u_n)| = \beta_n^{-1/2}$ with $\beta_n := \Theta'(2\pi n^2) = \log n + c + O(n^{-4})$; $|G_{n,\sigma}(\Theta(T))| = \Theta'(T)^{-1/2}$. So the boundary is bounded by $2\beta_n^{-1/2}/\lambda$.

Monotonicity of $G_{n,\sigma}$. $G_{n,\sigma}(u) = \Theta'(t(u))^{-1/2}$; since $\Theta'(t) = \vartheta'(t) + c$ is strictly increasing on $[2\pi n^2, \infty)$ (Lemma 1, $\vartheta'' > 0$ on $(0, \infty)$), $G_{n,\sigma}$ is strictly decreasing on $[u_n, \infty)$. Hence $G'_{n,\sigma} < 0$ and $|G'_{n,\sigma}| = -G'_{n,\sigma}$, giving

$$\int_{u_n}^{\infty} |G'_{n,\sigma}(u)| du = - \int_{u_n}^{\infty} G'_{n,\sigma}(u) du = G_{n,\sigma}(u_n) - \lim_{u \rightarrow \infty} G_{n,\sigma}(u) = \beta_n^{-1/2}. \quad (22)$$

Second-derivative bound. Differentiating (20) using $dt/du = 1/\Theta'(t)$,

$$\Psi''_{n,\sigma,\mu}(u) = \sigma \frac{d\omega_n(t)}{dt} \cdot \frac{dt}{du} = \sigma \cdot \frac{\vartheta''(t)(c + \log n)}{(\vartheta'(t) + c)^2} \cdot \frac{1}{\Theta'(t)} = \sigma \frac{\vartheta''(t)(c + \log n)}{\Theta'(t)^3}, \quad (23)$$

where the first factor is $d\omega_n/dt = \vartheta''(t)(c + \log n)/\Theta'(t)^2$ by quotient rule on $\omega_n(t) = (\vartheta'(t) - \log n)/(\vartheta'(t) + c)$. Therefore

$$\int_{u_n}^{\infty} |\Psi''_{n,\sigma,\mu}(u)| du = (c + \log n) \int_{2\pi n^2}^{\infty} \frac{\vartheta''(t)}{\Theta'(t)^2} dt = (c + \log n) \beta_n^{-1}, \quad (24)$$

using the substitution $u = \Theta(t)$, $du = \Theta'(t) dt$, and telescoping.

Tail. The integrand of the tail in (21) is dominated by $(|G'_{n,\sigma}| + |G_{n,\sigma}| |\Psi''_{n,\sigma,\mu}|/|\Psi'|)/|\Psi'|$; together with $|G_{n,\sigma}| \leq \beta_n^{-1/2}$, $|\Psi'| \geq \lambda$, and (22)–(24),

$$\left| \int_{u_n}^{\Theta(T)} \frac{d}{du} \left(\frac{G_{n,\sigma}}{i\Psi'_{n,\sigma,\mu}} \right) e^{i\Psi} du \right| \leq \frac{1}{\lambda} \beta_n^{-1/2} + \frac{\beta_n^{-1/2}(c + \log n)\beta_n^{-1}}{\lambda^2} = O_\lambda(\beta_n^{-1/2}), \quad (25)$$

since $(c + \log n)/\beta_n = (c + \log n)/(\log n + c + O(n^{-4})) \rightarrow 1$. Hence

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq C(c, |\mu|) \beta_n^{-1/2}, \quad (26)$$

uniform in T, n, σ , for $|\mu| > 1$.

Remainder $\mathcal{R}(T, \mu)$. The Riemann–Siegel remainder contributes

$$\mathcal{R}(T, \mu) = \frac{1}{2\pi} \int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt.$$

Write $R(t) = |R(t)|e^{i\alpha(t)}$; by Lemma 12, $|R(t)| = O(t^{-1/4})$ and $\alpha'(t) = O(t^{-1/2})$. The total phase is $\phi(t) := \alpha(t) - \mu\Theta(t)$; for t large, $\phi'(t) = O(t^{-1/2}) - \mu\Theta'(t) = -\mu\Theta'(t)(1 + o(1))$, so $|\phi'(t)| \geq |\mu|\Theta'(t)/2 \rightarrow \infty$. One integration by parts (pulling $|R(t)|\sqrt{\Theta'(t)}$ out of $e^{i\phi}$) gives

$$|\mathcal{R}(T, \mu)| \leq \frac{1}{2\pi} \left[\frac{|R(t)|\sqrt{\Theta'(t)}}{|\phi'(t)|} \right]_0^T + \frac{1}{2\pi} \int_0^T \left| \frac{d}{dt} \frac{|R(t)|\sqrt{\Theta'(t)}}{\phi'(t)} \right| dt.$$

Boundary: $|R(T)|\sqrt{\Theta'(T)}/|\phi'(T)| = O(T^{-1/4}\sqrt{\log T}/\log T) = O(T^{-1/4}/\sqrt{\log T}) = o(1)$. Tail: $|R|\sqrt{\Theta'}/|\phi'| = O(t^{-1/4}\sqrt{\log t}/\log t) = O(t^{-1/4}/\sqrt{\log t})$, with t -derivative $O(t^{-5/4})$; integrating gives $O(1)$. Hence $|\mathcal{R}(T, \mu)| \leq C_R(|\mu|)$ uniformly in T .

Assembly. From (17), (26), and the remainder bound,

$$|K_T(\mu)| \leq \frac{C(c, |\mu|)}{2\pi} \sum_{\sigma} \sum_{n=1}^{N(T)} n^{-1/2} \beta_n^{-1/2} + C_R(|\mu|).$$

Since $\beta_n \sim \log n$, $n^{-1/2} \beta_n^{-1/2} \sim n^{-1/2}(\log n)^{-1/2}$, and $\sum_{n \leq N(T)} n^{-1/2}(\log n)^{-1/2} = O(N(T)^{1/2}/\sqrt{\log N(T)}) = O(T^{1/4}/\sqrt{\log T})$. With $\Theta(T) \sim \frac{1}{2}T \log T$,

$$\frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{T^{1/2}/\log T}{T \log T}\right) = O\left(\frac{1}{T^{1/2}(\log T)^2}\right) \rightarrow 0, \quad (27)$$

so $S(\mu) = 0$ for $|\mu| > 1$. □

Corollary 14 (Support of μ_Y). $\text{supp}(\mu_Y) \subseteq [-1, 1]$.

Proof. By the Wiener–Khinchin identification for time-average-WSS functions, for any bounded Borel set $E \subset \mathbb{R}$,

$$\mu_Y(E) = \lim_{T \rightarrow \infty} \frac{2\pi}{\Theta(T)} \int_E |K_T(\mu)|^2 d\mu, \quad (28)$$

(Wiener, *Generalized Harmonic Analysis*, Acta Math. 55 (1930); Doob, *Stochastic Processes*, Ch. X §3). Take $E \subset \{|\mu| > 1\}$ bounded. From the proof of Theorem 13, $|K_T(\mu)|^2/\Theta(T) \leq C(c, |\mu|)^2/(T^{1/2}(\log T)^2) \rightarrow 0$ pointwise on E , with the integrand dominated by an $L^\infty(E)$ function uniformly in T . Dominated convergence on the bounded set E gives $\mu_Y(E) = 0$. Since $\{|\mu| > 1\}$ is the countable union of bounded sets $E_k := \{k < |\mu| \leq k+1\}$ for $k \geq 1$, countable additivity gives $\mu_Y(\{|\mu| > 1\}) = 0$, i.e., $\text{supp}(\mu_Y) \subseteq [-1, 1]$. □

4 Laguerre–Pólya Membership and Reality of Zeros of Y

Theorem 15 (Paley–Wiener extension). Y extends to an entire function of exponential type ≤ 1 satisfying $|Y(z)| \leq \mu_Y(\mathbb{R})^{1/2} e^{|\Im z|}$ on $\mathbb{C}$.

Proof. Y is time-average wide-sense stationary (Theorem 8) with covariance $c_Y(\eta) = \int e^{in\xi} d\mu_Y(\xi)$ (Corollary 9) and $\text{supp}(\mu_Y) \subseteq [-1, 1]$ (Corollary 14). The Paley–Wiener theorem for wide-sense stationary functions with compactly supported spectral measure (Yaglom, *Correlation Theory of Stationary and Related Random Functions*, Vol. I, Thm. 4.7.3; Bochner–Phillips, *Ann. Math.* 43 (1942)) gives that Y has an entire extension of exponential type ≤ 1 with the stated bound. □

Theorem 16 (Akhiezer Laguerre–Pólya criterion). *If F is a real-valued entire function of exponential type $\leq \tau$ whose autocorrelation measure (the Fourier transform of $|F|^2$ in the tempered-distribution sense) is a non-negative finite Borel measure supported in $[-2\tau, 2\tau]$, then F belongs to the Laguerre–Pólya class; all complex zeros of F are real.*

Proof. Akhiezer, *Theory of Approximation*, §V.4; Levin, *Lectures on Entire Functions*, Lecture 16 Thm. 3. $\square$

Theorem 17 (Reality of zeros of Y). *All zeros of the entire extension of Y are real.*

Proof. Y is real on $\mathbb{R}$ (Lemma 5) and entire of exponential type ≤ 1 (Theorem 15). The autocorrelation of Y is c_Y (Theorem 8), whose Fourier-transform measure is μ_Y (Corollary 9), non-negative and finite with $\text{supp}(\mu_Y) \subseteq [-1, 1] \subseteq [-2, 2]$ (Corollary 14). Apply Theorem 16 with $\tau = 1$. $\square$

5 Transfer to the Critical Line

Theorem 18 (Zero-set bijection via unitary warping). *$t_0 \in \mathbb{R}$ is a zero of Z of order m if and only if $\Theta(t_0) \in \mathbb{R}$ is a zero of Y of order m .*

Proof. $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ on $\mathbb{R}$ (Lemma 1). $Z(t_0) = 0 \iff Y(\Theta(t_0)) = 0$. For multiplicity, if Y has zero of order m at $s_0 = \Theta(t_0)$, $Y(s) = (s - s_0)^m g(s)$ with entire g , $g(s_0) \neq 0$. Substitute $s = \Theta(t)$: $\Theta(t) - \Theta(t_0) = (t - t_0)h(t)$ with $h(t) = \int_0^1 \Theta'(t_0 + u(t - t_0)) du$ and $h(t_0) = \Theta'(t_0) > 0$. Hence $Z(t) = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}$, whose trailing factor is continuous and nonzero at t_0 , giving zero of order exactly m . Converse by symmetry via Θ^{-1} . $\square$

Corollary 19 (Reality of zeros of Z). *Every zero of Z in $\mathbb{C}$ lies on $\mathbb{R}$: since Y has only real zeros (Theorem 17) and Θ is a real-analytic bijection $\mathbb{R} \rightarrow \mathbb{R}$ (Lemma 1), $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$, and the zero-set bijection (Theorem 18) transports reality back through Θ^{-1} .*

Theorem 20 (Zeros of ζ on the critical line). *For every $t_0 \in \mathbb{R}$, $\zeta(\frac{1}{2} + it_0) = 0$ if and only if $Y(\Theta(t_0)) = 0$, with equal multiplicities. Consequently, the zero set of ζ on the critical line $\{\text{Re } s = \frac{1}{2}\}$ is in multiplicity-preserving bijection with the real zero set of the entire function Y of exponential type ≤ 1 , under the map $t_0 \mapsto \frac{1}{2} + it_0$ composed with Θ . All nontrivial zeros of ζ satisfy $\text{Re } \rho = \frac{1}{2}$.*

Proof. For $t_0 \in \mathbb{R}$, $|e^{i\vartheta(t_0)}| = 1$, so $\zeta(\frac{1}{2} + it_0) = e^{-i\vartheta(t_0)} Z(t_0) = 0 \iff Z(t_0) = 0$, with matching multiplicities. Theorem 18 gives $Z(t_0) = 0 \iff Y(\Theta(t_0)) = 0$ with matching multiplicities. Theorem 17 gives all zeros of Y real, hence by Corollary 19 all zeros of Z are real. An off-critical-line zero $\rho = \beta + i\gamma$ of ζ with $\beta \neq \frac{1}{2}$ would correspond, via the holomorphic extension of $Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$ to the strip $|\Im t| < \frac{1}{2}$, to a zero of Z at $t = -i(\rho - \frac{1}{2}) = \gamma + i(\frac{1}{2} - \beta)$, which is non-real since $\beta \neq \frac{1}{2}$. Reality of all zeros of Z forces $\beta = \frac{1}{2}$; therefore every nontrivial zero of ζ lies on $\{\text{Re } s = \frac{1}{2}\}$. $\square$

References

- [1] N. I. Akhiezer, *Theory of Approximation*, Ungar, 1956.
- [2] S. Bochner and R. S. Phillips, Absolutely convergent Fourier expansions for non-commutative normed rings, *Ann. Math.* (2) **43** (1942), 409–418.

- [3] M. Clerc and S. Mallat, Estimating deformations of stationary processes, *Ann. Statist.* **31** (2003), no. 6, 1772–1821.
- [4] J. L. Doob, *Stochastic Processes*, Wiley, 1953.
- [5] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.
- [6] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Georg-August-Universität Göttingen, 1979.
- [7] G. H. Hardy and J. E. Littlewood, Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes, *Acta Math.* **41** (1918), 119–196.
- [8] B. Ya. Levin, *Lectures on Entire Functions*, Transl. Math. Monogr. **150**, Amer. Math. Soc., 1996.
- [9] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974.
- [10] H. Omer and B. Torresani, Time-frequency and time-scale analysis of deformed stationary processes, with application to non-stationary sound modeling, HAL preprint hal-01094835 (2017).
- [11] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, 1986.
- [12] N. Wiener, Generalized harmonic analysis, *Acta Math.* **55** (1930), 117–258.
- [13] A. M. Yaglom, *Correlation Theory of Stationary and Related Random Functions*, Vol. I, Springer, 1987.